

JUSTIN GOTTLIEB

MANUFACTURING SYSTEMS & TEST AUTOMATION ENGINEER

Strongsville, Ohio | 216-296-6690 | justingottliebengineering.github.io

PROFESSIONAL SUMMARY

Manufacturing systems and test automation engineer and U.S. Army veteran with nearly six years of progressive experience at Bird, advancing from RF Test Technician to Manufacturing Test Engineer II. Designs and deploys technician-facing production systems that modernize legacy programming, test, maintenance, and data workflows. Combines manufacturing process analysis, software architecture, embedded-device programming, instrument integration, SQL traceability, and automated testing to improve repeatability, product reliability, production scalability, and delivery performance.

SELECTED ENGINEERING IMPACT

PRODUCTION MODERNIZATION	RELIABILITY & TRACEABILITY	SCALABILITY & ADOPTION
Unified fragmented programming and test practices into controlled technician workflows.	Built systems that preserve configuration, execution, measurement, and maintenance history.	Designed repeatable platforms for new products, higher volume, and reduced reliance on tribal knowledge.

CORE EXPERTISE

Manufacturing Systems Architecture • PCBA Test Automation • RF Calibration & Troubleshooting • Technician Workflow Design • Embedded Programming • Instrument Integration • SQL Traceability • AOI/SPI & SMT Programming • Manufacturing Analytics • Preventive Maintenance Systems • Git-Based Release Control • AI-Assisted Engineering Development

PROFESSIONAL EXPERIENCE

Manufacturing Test Engineer II | Bird Technologies - Solon, OH **Jan 2023 - Present**

- Led modernization of legacy PCBA production practices by designing and deploying a centralized programming, functional-test, and traceability platform that replaces fragmented technician-dependent procedures with a controlled, database-driven workflow.
- Architected technician-facing applications in Python, Flask, Django, PySide6, SQL/SQLite/SQL Server, HTML/CSS/JavaScript, and Git to standardize firmware control, test execution, work instructions, preventive maintenance, stencil lookup, reporting, and production-data retrieval.
- Integrated automated firmware staging, embedded-device programming, live execution status, rerun and abort handling, failure recovery, equipment selection, and persistent result records across Windows and Linux environments.
- Presented the manufacturing-platform modernization initiative during the annual board of directors plant review, framing the system as a scalable foundation for expected volume growth, improved product reliability, and stronger on-time delivery performance.
- Develop and sustain embedded programming workflows using JTAG, ISP, DFU, SWD, UART/RS-232, USB, Ethernet/TFTP, Atmel-ICE, USB-Blaster, OpenOCD, AVRDUDE, dfu-programmer, Quartus Programmer, and vendor-specific utilities.
- Lead troubleshooting and integration of legacy and modern electronic test equipment using GPIB/VISA, SCPI, serial protocols, USB, TCP/IP, frequency counters, vector network analyzers, signal generators, RF power meters, and custom fixtures.
- Program and maintain Saki 3D AOI/SPI and ePm workflows; resolve footprint, pad, rotation, panelization, bottom-side placement, and component-data problems across ODB++, Gerber, BOM/CSV, JSON, and manufacturing-source files.
- Create and validate ASM SMT placement programs from ODB++ data, including panel and PCB step structures, component-layer mapping, fiducials, board profiles, and top/bottom placement integrity.
- Designed an ISO 9001-oriented preventive-maintenance management system with equipment QR access, revision-controlled procedures, scheduled work orders, technician evidence capture, role-based review, notifications, and auditable equipment history.
- Use manufacturing, QIS, test, and maintenance data to automate product configuration, calibration requirements, historical-result retrieval, yield analysis, defect prioritization, and engineering decision support.
- Partner with technicians, manufacturing engineering, quality, IT, leadership, and equipment vendors to translate manual or error-prone operations into controlled, maintainable systems and drive user adoption.
- Use generative AI to accelerate research, debugging, documentation, test-case development, and workflow refinement while independently validating outputs and retaining final engineering ownership.

RF Test Technician | Bird Technologies - Solon, OH **Dec 2020 - Jan 2023**

- Tested, tuned, repaired, and calibrated RF products and systems for power accuracy, insertion loss, return loss, impedance, gain, and spectral performance.
- Performed calibration and verification of calorimeters, RF power sensors, pulse sensors, BDS systems, filters, boosters, loads, and related electromechanical assemblies.
- Diagnosed failures to component, cable, fixture, firmware, or process level using vector network analyzers, signal generators, spectrum analyzers, oscilloscopes, frequency counters, and RF power meters.

Low Voltage Technician | GTS Communications - Strongsville, OH **Apr 2020 - Dec 2020**

- Installed, terminated, tested, and serviced CAT5/6 cabling, patch panels, wireless access points, surveillance cameras, and related commercial low-voltage infrastructure.

Telecommunications Manager / Sergeant | United States Army - Fort Drum, NY / Afghanistan **Oct 2008 - Oct 2013**

- Led communications readiness, training, maintenance, and accountability for 106 enlisted Soldiers, 25 officers, and mission-critical radio, satellite, computer, and battlefield command systems valued at more than \$50 million.
- Created preventive-maintenance, inventory-control, shortage-tracking, and equipment-readiness systems that achieved full accountability and operational availability.
- Developed SOPs, training plans, databases, spreadsheets, and command briefings; trained more than 1,000 Soldiers in radio operation and maintenance.
- Deployed to Afghanistan supporting communications operations and infrastructure rewiring for Regional Command South.

ADDITIONAL LEADERSHIP EXPERIENCE

Retail Operations Leadership - Party City, Hobby Lobby, Halloween City, Sears/Kmart, and Harbor Freight Tools | 2006 - 2020

Managed staffing, training, payroll, inventory, P&L performance, loss prevention, merchandising, and store operations for teams ranging from 10 to more than 150 employees, demonstrating long-term leadership, coaching, budgeting, and operational discipline.

EDUCATION & CREDENTIALS

Bachelor of Science, Information Technology (E-Commerce) - American Military University - Summa Cum Laude, GPA 4.00, 2016

Associate of Science, Web Publishing - American Military University - With Honors, 2015

CompTIA Network+ - 2013 | OSHA 10-Hour General Industry Safety and Health - 2025

U.S. Army Signal Support Systems Specialist - Honor Graduate | Warrior Leader Course - Commandant's List

TECHNICAL TOOLS

Software & Data

Python, R, Flask, Django, PySide6, HTML, CSS, JavaScript, SQL, SQLite, SQL Server, Git/GitHub, PowerShell, Windows Batch, Quarto, tidyverse, ggplot2, testthat

Manufacturing & Inspection

Saki 3D AOI/SPI, ePm, ASM SMT programming, ODB++, Gerber, BOM/CSV, JSON, Seica flying probe workflows, preventive-maintenance systems

Embedded & Interfaces

JTAG, ISP, DFU, SWD, UART/RS-232, USB, Ethernet/TFTP, GPIB/VISA, SCPI, OpenOCD, AVRDUDE, dfu-programmer, Quartus Programmer, Atmel-ICE, USB-Blaster

RF & Electronic Test

Vector network analyzers, signal generators, spectrum analyzers, oscilloscopes, frequency counters, RF power meters, custom fixtures

SELECTED HONORS

Army Commendation Medal (2) | Army Good Conduct Medal | Afghanistan Campaign Medal with Two Campaign Stars | NATO Medal